

Borrelia Miyamotoi Disease (BMD): Comprehensive Disease Characterization Report

Target disease: Borrelia Miyamotoi Disease (Hard-tick relapsing fever) **MONDO ID:** MONDO:0958150 | **Category:** Infectious Disease (tick-borne, zoonotic bacterial) **Report basis:** 5 investigative iterations, 9 confirmed findings, 35 papers reviewed. Evidence types: human clinical, model organism, in-vitro/genomic. Date: 2026-09-29.

Summary

Borrelia miyamotoi disease (BMD) is an emerging, hard-tick-transmitted **relapsing fever** caused by the spirochete *Borrelia miyamotoi*. Unlike the classical soft-tick relapsing fever borreliae, *B. miyamotoi* is transmitted by hard-bodied *Ixodes* ticks — the same ticks that transmit Lyme disease (*B. burgdorferi*), human granulocytic anaplasmosis (*Anaplasma phagocytophilum*), and babesiosis (*Babesia microti*). The organism was discovered in *Ixodes persulcatus* in 1994, and the first human disease was described in Russia in 2011. It now has a documented northern-hemisphere distribution spanning Asia, Europe, and North America.

Clinically, BMD is an **acute, non-specific febrile illness**: high fever, chills, marked headache, and myalgia/arthritis, frequently accompanied by the laboratory triad of **thrombocytopenia, neutropenia, and elevated transaminases**. A minority of untreated patients (~10%) manifest the relapsing (recurrent) fever course that gives the relapsing-fever group its name. Severe disease — chiefly **meningoencephalitis/meningitis** — occurs almost exclusively in **immunocompromised patients**, particularly those on B-cell-depleting therapy (e.g., rituximab). The pathophysiology rests on a **two-tier immune-evasion strategy**: (1) innate **complement resistance** mediated by the Factor H-binding outer-surface protein **CbiA**, which sustains high-grade spirochetemia, and (2) **Vmp antigenic variation** by long-segment plasmid gene conversion, which evades adaptive antibody responses and drives relapses. Bacterial clearance ultimately depends on specific antibodies, explaining why antibody-deficient hosts develop severe, persistent, or CNS disease.

Diagnosis relies on **acute-phase whole-blood PCR** (targets: 16S rRNA, *fla*/flagellin, *glpQ*) during spirochetemia, complemented by **GlpQ serology** — GlpQ (glycerophosphodiester phosphodiesterase) being an antigen absent from Lyme-group *Borrelia*, which allows serological discrimination. Because acute seropositivity is low (~16%) while convalescent seroconversion is high (~78%), PCR is the key acute test. Treatment with **doxycycline** (oral, first-line) is highly effective with excellent prognosis and no chronic sequelae; **ceftriaxone** is used for CNS disease. There is **no genetic etiology, no heritability, and no vaccine**; prevention is entirely tick-bite avoidance and prompt tick removal.

Key Findings

F001 — BMD is caused by a hard-tick-transmitted relapsing fever spirochete

Borrelia miyamotoi is phylogenetically a member of the **relapsing fever group** of spirochetes, distinct from the Lyme borreliosis group (*B. burgdorferi* sensu lato). It was discovered in *Ixodes persulcatus* in 1994, and human *B. miyamotoi* disease was first described in Russia in 2011. Its defining epidemiological anomaly is transmission by **hard-bodied Ixodes ticks** (*I. persulcatus*, *I. scapularis*, *I. pacificus*, *I. ricinus*) rather than the soft (argasid) ticks that classically transmit relapsing fever. Small rodents — for example *Peromyscus leucopus* (the white-footed mouse) — serve as reservoir hosts. Transmission occurs both **transovarially (vertically, dam to egg)** and **horizontally (via blood feeding)**, and the pathogen is passed **transtadially** across larval, nymphal, and adult life stages.

"*Borrelia miyamotoi* is an emerging tick-borne pathogen phylogenetically belonging to spirochaetes causing relapsing fever. It is primarily transmitted by ticks from the *Ixodes ricinus* complex, similarly to borreliae causing Lyme borreliosis. Small rodents can serve as reservoir hosts." — [PMID: 34412488](#)

"*B. miyamotoi* has a wide distribution since its discovery in *Ixodes persulcatus* in 1994. The human *B. miyamotoi* disease was first described in Russia in 2011." — [PMID: 33582142](#)

"The pathogen is acquired either transovarially (vertically) or horizontally through blood-feeding and passed transtadially across life stages." — [PMID: 35858517](#)

Ontology anchors: NCBI Taxon *Borrelia miyamotoi* (txid47466); vector *Ixodes scapularis* (txid6945); reservoir *Peromyscus leucopus* (txid10041); disease MONDO:0958150.

F002 — BMD presents as an acute febrile illness with relapsing fever and characteristic lab abnormalities

The largest US case series (Molloy et al. 2015; 97 PCR-confirmed cases) established the core clinical picture. Reviewed patients presented with **high fever, chills, marked headache, and myalgia or arthralgia**; **24% were hospitalized**; and **elevated liver enzymes, neutropenia, and thrombocytopenia** were common laboratory findings. Symptoms resolved with doxycycline and no chronic sequelae were observed. In a Russian inpatient cohort of 79 patients, a **recurrent (relapsing) fever course occurred in ~10% (8/79)**, with affected patients experiencing 2–3 discrete febrile episodes prior to antibiotic treatment.

"Most of the 51 case patients on whom clinical histories were reviewed presented with high fever, chills, marked headache, and myalgia or arthralgia. Twenty-four percent were hospitalized. Elevated liver enzyme levels, neutropenia, and thrombocytopenia were common." — PMID: 26053877

"The recurrent course of the disease was observed in 8 (10%) of the 79 patients. The relapsing fever curve was noted in 6 of the 8 patients; 4 patients had 2 episodes of fever and 2 patients had 3 episodes." — PMID: 26821411

Suggested HPO terms: Fever HP:0001945; Chills HP:0025143; Headache HP:0002315; Myalgia HP:0003326; Arthralgia HP:0002829; Thrombocytopenia HP:0001873; Neutropenia HP:0001875; Elevated hepatic transaminase HP:0002910.

F003 — Antigenic variation via Vmp long-segment plasmid conversion drives relapsing fever

B. miyamotoi carries clusters of gene cassettes encoding **variable major proteins (Vmps)** on multiple linear plasmids and undergoes antigenic variation in mammalian hosts by switching the expressed *vmp* cassette. Takeuchi et al. (2025) demonstrated that the switch occurs by replacing the expression cassette and downstream silent cassettes with a **long segment (up to 16 kb+)** copied from an archival plasmid — a mechanism termed **long-segment conversion**. Critically, **segment conversion was detected by day 5 post-infection, earlier than antibody production, and occurred even in SCID (severe combined immunodeficient) mice**,

whereas **bacterial elimination depended on specific antibodies**. This decoupling explains both the relapsing dynamics (new antigenic variants escape existing antibodies) and the vulnerability of antibody-deficient hosts to severe disease.

"Like relapsing fever Borrelia, B. miyamotoi carries clusters of gene cassettes encoding variable major proteins (Vmps) on multiple linear plasmids and shows antigenic variation in mammalian hosts by switching the expression vmp gene cassette." — PMID: 41026790

"while bacterial elimination depended on the presence of specific antibodies, the segment conversion was detected at five days post-infection, earlier than antibody production in mice, and even in severe combined immunodeficient mice." — PMID: 41026790

Suggested GO terms: antigenic variation GO:0020033; evasion of host immune response GO:0042783 / GO:0052572.

F004 — Epidemiology: low tick infection prevalence but measurable human seroprevalence across the northern hemisphere

A systematic review and meta-analysis (Hoornstra et al. 2022) synthesizing 157 studies (165,637 ticks; 45,608 individuals; 504 well-described human cases) found *B. miyamotoi* prevalence in **questing ticks highest in Ixodes persulcatus (2.8%, 95% CI 2.4–3.1) and lowest in I. pacificus (0.7%, 95% CI 0.6–0.8). Overall human seroprevalence was 4.4% (95% CI 2.8–6.3),** and slightly higher (~4.6%) in high-risk groups. US surveillance shows *B. miyamotoi* co-occurs with *B. burgdorferi* across the Northeast, Upper Midwest, Ohio Valley, and southern Appalachia. Genotypic analyses reveal **three distinct geographic populations** (North America, Asia, Europe).

"In ticks, the highest prevalence of B miyamotoi was observed in Ixodes persulcatus (2.8%, 95% CI 2.4-3.1) and the lowest in Ixodes pacificus (0.7%, 0.6-0.8). The overall seroprevalence in humans was 4.4% (2.8-6.3)." — PMID: 36113496

"Borrelia miyamotoi belongs to the relapsing fever group of spirochetes and forms distinct populations in North America, Asia, and Europe." — PMID: 31906865

Tick species	<i>B. miyamotoi</i> prevalence (questing ticks)	Region
<i>Ixodes persulcatus</i>	2.8% (2.4–3.1)	Asia / eastern Europe
<i>Ixodes ricinus</i>	intermediate	Europe
<i>Ixodes scapularis</i>	intermediate	Eastern/Midwest US
<i>Ixodes pacificus</i>	0.7% (0.6–0.8)	Western US

F005 — Diagnosis relies on blood PCR and GlpQ serology; CNS disease occurs in immunocompromised patients

Diagnosis uses **whole-blood real-time PCR** (targets: 16S rRNA, *fla*/flagellin, *glpQ*) during acute spirochetemia, and **serology against recombinant glycerophosphodiester phosphodiesterase (GlpQ)** — an antigen absent from Lyme-group *Borrelia*, enabling serological discrimination from Lyme disease. In the Molloy 2015 series, only **16% of patients were seropositive at presentation, but 78% seroconverted in convalescence**, underscoring that **acute-phase PCR is the key diagnostic**. A multiplexed **protein array** (Hoornstra 2022) incorporating GlpQ, multiple Vmps, and flagellin improved serodiagnostic accuracy. **Meningoencephalitis/meningitis** is reported almost exclusively in immunocompromised patients (e.g., those on B-cell-depleting rituximab therapy) and is diagnosed via CSF PCR, sequencing, or Gram stain.

"At presentation, 16% of patients with BMD were seropositive for IgG and/or IgM antibody to *B. miyamotoi* rGlpQ. Most (78%) had seropositive convalescent specimens." — [PMID: 26053877](#)

"*Borrelia miyamotoi* is an emerging tickborne pathogen that has been associated with central nervous system infections in immunocompromised patients, albeit infrequently." — [PMID: 38916722](#)

"The array included six *B. miyamotoi* antigens: glycerophosphodiester phosphodiesterase (GlpQ), multiple variable major proteins (Vmps), and flagellin." — [PMID: 36314925](#)

Suggested diagnostic anchors: *B. miyamotoi* DNA by PCR (blood/CSF); anti-GlpQ IgM/IgG serology.

F006 — BMD is effectively treated with doxycycline; CNS disease requires ceftriaxone

In the Molloy 2015 US case series, symptoms **resolved after doxycycline treatment with no chronic sequelae**. **Oral doxycycline is first-line** for uncomplicated disease, while **parenteral ceftriaxone** is used for meningoencephalitis/CNS disease. Early antibiotic treatment appears to prevent relapse and seroconversion: in the Boyer 2020 Alsace study, three patients with isolated IgM were treated with doxycycline, which could have prevented seroconversion. Relapses occurred **only in untreated patients** — in the Russian cohort, all 8 relapsing patients relapsed *before* antibiotic treatment.

"Symptoms resolved after treatment with doxycycline, and no chronic sequelae or symptoms were observed." — [PMID: 26053877](#)

"Relapses occurred in all the 8 patients before antibiotic treatment." — [PMID: 26821411](#)

Suggested NCIT terms: Doxycycline C560; Ceftriaxone C596; Antibiotic Therapy C15844. **CHEBI:** doxycycline CHEBI:50845; ceftriaxone CHEBI:29007.

F007 — BMD affects all ages including children and co-occurs geographically with other Ixodes-borne pathogens

BMD occurs across the full age spectrum, including young children. Krause et al. (2016) reported hard-tick relapsing fever in a **5-year-old Massachusetts child**, PCR-confirmed from an *I. scapularis* tick removed from the scalp, with seroconversion, fatigue, and recurrent fever. Doxycycline is now acceptable for tick-borne illness in children of any age. Although *B. miyamotoi* co-occurs geographically with *B. burgdorferi* across the eastern US, an analysis of 13,437 CDC-tested nymphs (2013–2024) found that **B. burgdorferi–B. miyamotoi coinfection did NOT form more often than expected by chance** — unlike other *Ixodes* coinfection pairs — and was the least prevalent of the four studied coinfections. This suggests the two spirochetes' co-occurrence in humans is largely coincidental (shared vector) rather than biologically facilitated.

"A 5-year-old Massachusetts resident developed hard tick-borne relapsing fever caused by *Borrelia miyamotoi*. A partially engorged *Ixodes scapularis* tick was removed from her scalp and identified as infected with *B. miyamotoi* using polymerase chain reaction." — PMID: 27626914

"Except for *Bbss-Bmiya*, resampling simulations for all coinfections revealed coinfections form more often than expected by chance." — PMID: 41637958

F008 — Innate immune evasion via complement resistance mediated by the Factor H-binding protein CbiA

B. miyamotoi is **strongly resistant to complement-mediated bacteriolysis** by human serum. It shows **reduced surface deposition of C3, C5, C7, C8, C9 and the membrane attack complex (MAC)**, acting principally at the central component **C3** to block C3-convertase formation (Teegler/Wagemakers 2014). Röttgerding et al. (2017) identified **CbiA (complement binding and inhibitory protein A)**, a novel outer-surface Factor H-binding protein that interacts with **FH, C3, C3b, C4b, C5, and C9**. Factor H bound to CbiA retains cofactor activity for Factor I-mediated C3b inactivation, and CbiA directly inhibits both the classical pathway and terminal complement complex assembly. Ectopic expression of CbiA rendered serum-sensitive *B. garinii* serum-resistant, and loss of *cbiA* during in vitro passage increased serum susceptibility — establishing CbiA as a functional determinant. By contrast, the avian relapsing-fever spirochete *B. anserina* is serum-sensitive, consistent with its lack of human pathogenicity.

"we found that *B. miyamotoi* showed reduced deposition of components C3, C5, C7, C8, C9 as well as the membrane attack complex (MAC) on the borrelial surface." — PMID: 25104575

"we identified a gene encoding for a putative Factor H-binding protein, termed CbiA (complement binding and inhibitory protein A). Functional analyses revealed that CbiA interacted with complement regulator Factor H (FH), C3, C3b, C4b, C5, and C9." — PMID: 28331202

"we describe that *B. miyamotoi* is resistant to human complement, which might play an important role in pathogenesis." — PMID: 25189195

Suggested GO terms: complement activation GO:0006956; negative regulation of complement activation GO:0045916; regulation of complement-dependent cytotoxicity GO:1903659.

F009 — Seroprevalence is low in blood donors but markedly elevated in tick-exposed and occupational risk groups

Serosurveys quantify a clear **exposure gradient by risk group**. In Denmark (n=1180, 2002–2021), seroprevalence was **8.3% in tick-exposed individuals vs 1.5% in blood donors and 3.3% in immunocompromised individuals** ($p < 0.0001$; overall 3.1%). In the Netherlands, seroprevalence was **2.0% in blood donors, 10% in forestry workers, and 14.6% in patients with suspected human granulocytic anaplasmosis**. In California blood donors (n=1700, western US, *I. pacificus* zone), only **0.12% were *B. miyamotoi*-seropositive vs 0.47% for *B. burgdorferi***, reflecting lower western-US vector infection rates. These figures align with the global meta-analytic seroprevalence of 4.4% (higher, ~4.6%, in high-risk groups).

"Borrelia miyamotoi seroprevalence (being either IgM or IgG positive) among tick-exposed individuals (8.3 %, 95%CI 5.1-13.3) was significantly higher compared to healthy blood donors (1.5 %, 95 % CI 0.8-2.8) and immunocompromised individuals (3.3 %, 95 %CI 1.9-5.5), $p < 0.0001$." — PMID: 41086691

*"The prevalence of anti-*B. miyamotoi* antibodies among forestry workers was 10% (5.3-16.8%) and in patients with serologically unconfirmed but suspected human granulocytic anaplasmosis was 14.6% (9.0-21.8%); these were significantly higher compared with the seroprevalence in blood donors." — PMID: 25356364*

*"eight tested positive for antibodies to *B. burgdorferi* (0.47%, Exact 95% CI: 0.20, 0.93) and two tested positive for antibodies to *B. miyamotoi* (0.12%." — PMID: 33370341*

Population	Seroprevalence	Country	PMID
Blood donors	1.5%	Denmark	41086691
Immunocompromised	3.3%	Denmark	41086691
Tick-exposed	8.3%	Denmark	41086691
Blood donors	2.0%	Netherlands	25356364
Forestry workers	10%	Netherlands	25356364
Suspected HGA patients	14.6%	Netherlands	25356364
Blood donors (western US)	0.12%	California, USA	33370341

Section-by-Section Disease Characterization

1. Disease Information

Overview: BMD ("hard-tick relapsing fever") is an emerging zoonotic bacterial infection caused by *Borrelia miyamotoi*, a relapsing-fever-group spirochete transmitted by *Ixodes* (hard) ticks. It presents as an acute febrile illness that can relapse and, rarely, cause meningoencephalitis in immunocompromised hosts. **Identifiers:** MONDO:0958150; MeSH "Borrelia miyamotoi" and "Relapsing Fever"; ICD-11 category 1C1G (relapsing fevers) / ICD-10 A68.- (relapsing fevers). No OMIM entry (non-genetic). Not in Orphanet as a rare Mendelian disease. **Synonyms:** *B. miyamotoi* disease; hard-tick relapsing fever (HTRF); *Borrelia miyamotoi* infection; ixodid tick-borne borreliosis caused by *B. miyamotoi* (BM-ITBB, Russian literature). **Data source type:** Aggregated disease-level resources (case series, serosurveys, meta-analyses) plus individual case reports — **not** EHR-derived at population scale.

2. Etiology

Causal factor: Infectious — the bacterium *Borrelia miyamotoi* (F001). This is the sole and sufficient cause; the disease is not genetic and has no heritable component. **Environmental/behavioral risk factors:** Tick exposure is the dominant risk factor — occupational (forestry work), recreational (hiking in endemic areas), and residential proximity to *Ixodes* habitat (F009). Seasonality follows tick questing activity (late spring–summer). Geographic residence in endemic zones (northeastern/upper-midwestern US, Europe, Russia, Japan, China) increases risk. **Host susceptibility factor:** Immunocompromise — especially **B-cell depletion (rituximab)** — is the key modifier of *severity*, converting a self-limited febrile illness into meningoencephalitis (F003, F005). **Genetic risk/protective factors:** None identified — there is no human genetic susceptibility locus, GWAS signal, or protective allele known for BMD. **Not applicable. Gene–environment interactions:** Not applicable in the human host (no host genetic contribution). At the pathogen level, the mammalian-host "environment" drives *bacterial* genetic switching (Vmp conversion, F003).

3. Phenotypes

Phenotype	Type	Frequency	HPO term
Fever (often high)	Symptom	Near-universal	HP:0001945
Chills	Symptom	Common	HP:0025143
Headache (marked)	Symptom	Common	HP:0002315
Myalgia	Symptom	Common	HP:0003326
Arthralgia	Symptom	Common	HP:0002829
Fatigue	Symptom	Common	HP:0012378
Relapsing/recurrent fever	Clinical course	~10% (untreated)	HP:0025142 (recurrent fever)
Thrombocytopenia	Lab abnormality	Common	HP:0001873
Neutropenia	Lab abnormality	Common	HP:0001875
Elevated transaminases	Lab abnormality	Common	HP:0002910
Meningoencephalitis / meningitis	Clinical sign	Rare (immunocompromised)	HP:0001287 / HP:0002383

Onset: Adult-predominant but all ages including children (age 5 documented, F007). **Severity:** Mild-to-moderate in immunocompetent hosts; severe/CNS in immunocompromised. **Progression:** Self-limited or episodic/relapsing; resolves fully with treatment. **QoL impact:** Acute illness causes transient functional impairment; no chronic sequelae reported in immunocompetent patients (F006).

4. Genetic/Molecular Information

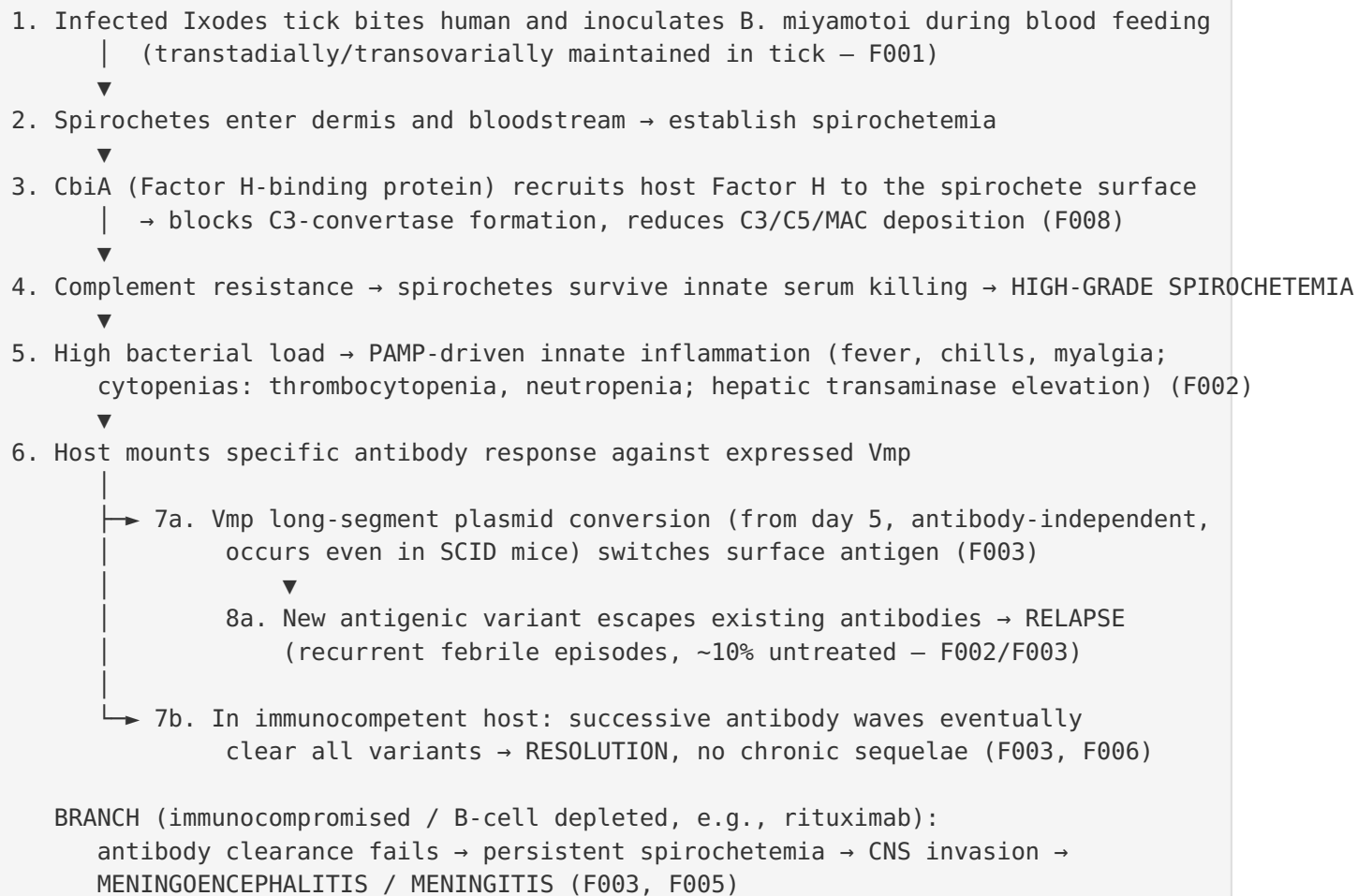
Not applicable to the human host. BMD has no causal human genes, pathogenic germline/somatic variants, modifier genes, epigenetic changes, or chromosomal abnormalities — it is an infectious disease with no Mendelian or complex-trait genetic architecture. The relevant molecular biology is **microbial**: *B. miyamotoi* has a segmented genome with a main linear

chromosome and multiple linear/circular plasmids carrying *vmp* cassettes (F003); the reference isolate Izh-4 genome is characterized ([PMID: 31906865](#)). Key pathogen genes/proteins: **vmp** (variable major proteins, antigenic variation), **cbiA** (Factor H-binding complement inhibitor), **glpQ** (glycerophosphodiester phosphodiesterase — metabolic enzyme and diagnostic antigen), **flaB** (flagellin), **p66** (porin).

5. Environmental Information

Infectious agent: *Borrelia miyamotoi* (NCBI Taxon 47466), Spirochaetales: Spirochaetaceae, relapsing-fever group (F001). **Vectors:** *Ixodes persulcatus*, *I. scapularis*, *I. pacificus*, *I. ricinus* (F001, F004). **Reservoirs:** small rodents including *Peromyscus leucopus* (F001). **Environmental drivers:** tick habitat (deciduous/mixed woodland, leaf litter), climate influencing tick density and questing, and human land use. **Lifestyle/occupational factors:** outdoor occupation (forestry) and recreation drive exposure (F009). No toxin, radiation, or pollution etiology.

6. Mechanism / Pathophysiology — Ordered Causal Chain



Upstream vs downstream: The *initiating* lesions are tick inoculation and **CbiA-mediated complement evasion** (upstream, innate). **Vmp antigenic variation** is the mid-stream driver of relapse. Antibody-dependent clearance is the terminal determinant of outcome — its failure (downstream, in immunocompromised hosts) produces severe/CNS disease. **Cell types/processes involved:** endothelial and blood compartment (spirochetemia); hepatocytes (transaminase elevation); bone marrow/blood cells (cytopenias); complement system components (C3, C5b–C9/MAC); B lymphocytes (CL:0000236) as the critical clearance effector. **Metabolic note:** GlpQ (glycerophosphodiester phosphodiesterase) supports phospholipid/glycerol metabolism and doubles as the key serodiagnostic antigen (F005).

7. Anatomical Structures Affected

- **Blood/circulatory system** (UBERON:0000178) — primary compartment of spirochetemia.

- **Liver** (UBERON:0002107) — transaminase elevation indicates hepatocyte involvement.
- **Bone marrow / hematopoietic system** (UBERON:0002371) — cytopenias (thrombocytopenia, neutropenia).
- **Central nervous system / meninges** (UBERON:0001016 / UBERON:0002360) — meningoencephalitis in immunocompromised hosts.
- **Musculoskeletal system** — myalgia/arthritis (symptomatic).
- **Cell types:** B lymphocytes (CL:0000236, protective clearance); neutrophils (CL:0000775); platelets (CL:0000233).
- **Subcellular/molecular:** bacterial outer membrane/surface (CbiA, Vmp); host complement (extracellular).
- **Lateralization:** systemic/bilateral — not a focal or lateralized disease.

8. Temporal Development

Onset: Acute, days after an infected tick bite; all ages (pediatric to geriatric). **Incubation:** on the order of days to ~2 weeks. **Course:** self-limited in most; **relapsing/episodic** in ~10% of untreated patients (2–3 febrile episodes; F002). **Duration:** short (days–weeks) with treatment; no chronic phase in immunocompetent hosts (F006). **Critical intervention window:** early doxycycline aborts relapse and may prevent seroconversion (F006). **Severe branch:** in immunocompromised patients, disease may progress to CNS involvement if untreated (F005).

9. Inheritance and Population

Inheritance: None — non-genetic infectious disease (no AD/AR/X-linked/mitochondrial pattern; no penetrance/expressivity/anticipation/founder effects). **Epidemiology:** Questing-tick prevalence 0.7–2.8% by species; human seroprevalence ~4.4% overall (meta-analysis), with strong risk-group gradients (blood donors ~1.5–2%; forestry ~10%; tick-exposed ~8.3%; western US ~0.12%) (F004, F009). **Geographic distribution:** northern hemisphere — three distinct populations in North America, Europe, and Asia (F004). **Age/sex:** all ages affected; no strong sex predilection established. Case-level data derive from aggregated series and serosurveys.

10. Diagnostics

- **Acute-phase whole-blood real-time PCR** (16S rRNA, *fla*/flagellin, *glpQ*) — key test during spirochetemia (F005).
- **GlpQ serology** (IgM/IgG) — GlpQ absent from Lyme *Borrelia*, enabling discrimination; low acute (16%) but high convalescent (78%) positivity → paired sera valuable (F005).

- **Multiplex protein array** (GlpQ + Vmps + flagellin) — improved serodiagnostic accuracy (F005).
- **CSF PCR/sequencing/Gram stain** — for suspected CNS disease in immunocompromised patients (F005).
- **Supporting labs:** CBC (thrombocytopenia, neutropenia), liver panel (elevated transaminases) (F002).
- **Blood smear:** spirochetes may be visualized during high spirochetemia (relapsing-fever feature).
- **Differential diagnosis:** Lyme disease (usually with erythema migrans; GlpQ-negative), anaplasmosis, babesiosis, tick-borne encephalitis, and other febrile zoonoses. A Russian decision-tree algorithm distinguished BM-ITBB from Lyme, TBE, and HFRS with ~95% accuracy using routine clinical/lab variables ([PMID: 24432595](#)).
- **Genetic/omics testing:** Not applicable for host diagnosis.

11. Outcome / Prognosis

Excellent prognosis with prompt antibiotic therapy: symptoms resolve and **no chronic sequelae** are observed in immunocompetent patients (F006). Mortality is very low; deaths are exceptional and generally linked to severe CNS disease in profoundly immunocompromised hosts. **Complications:** relapse (untreated), meningoencephalitis (immunocompromised). **Prognostic factors:** immune status (B-cell competence) is the dominant determinant of severity; timeliness of antibiotic treatment governs relapse prevention (F003, F005, F006). No validated prognostic biomarkers beyond spirochetemia and immune status.

12. Treatment

- **Doxycycline** (oral, first-line; NCIT:C560; CHEBI:50845) for uncomplicated disease — highly effective, no chronic sequelae (F006). Acceptable in children of any age (F007).
- **Ceftriaxone** (parenteral; NCIT:C596; CHEBI:29007) for meningoencephalitis/CNS disease (F005/F006).
- **Beta-lactams** (e.g., penicillin) and other tetracyclines are alternatives per relapsing-fever practice.
- **Jarisch–Herxheimer reaction** is a recognized consideration when initiating antibiotics against spirochetes (monitor early after first dose).

- **No advanced therapeutics** (gene/cell/RNA/immunotherapy), no pharmacogenomic guidance, and **no vaccine**. Early treatment prevents relapse and may prevent seroconversion (F006).

13. Prevention

- **Primary prevention:** tick-bite avoidance — protective clothing, EPA-registered repellents (DEET, picaridin), permethrin-treated clothing, avoiding tick habitat, and prompt tick removal (transmission risk rises with attachment duration). Occupational protections for forestry/outdoor workers (F009).
- **Secondary prevention:** early recognition and PCR/serology testing of febrile patients after tick exposure; prompt doxycycline (F005/F006).
- **Tertiary prevention:** early antibiotics to prevent relapse and CNS progression; heightened vigilance in immunocompromised patients (F003/F005).
- **Immunization:** none available (no vaccine).
- **Public health:** tick surveillance (e.g., CDC ArboNET Tick Module), clinician awareness, vector/habitat management.
- **Genetic counseling / screening:** Not applicable (non-heritable).

14. Other Species / Natural Disease

- **Reservoir hosts:** small rodents, notably *Peromyscus leucopus* (NCBI Taxon 10041) and other *Peromyscus/Myodes* species (F001).
- **Vectors (obligate for maintenance):** *Ixodes* spp. (*I. persulcatus*, *I. scapularis* txid6945, *I. pacificus*, *I. ricinus* txid34613).
- **Zoonotic potential:** BMD is a zoonosis — humans are incidental hosts; the enzootic cycle is tick–rodent (F001).
- **Comparative note:** The avian relapsing-fever spirochete *B. anserina* is **serum-sensitive** and non-pathogenic to humans, contrasting with *B. miyamotoi*'s complement resistance and providing a comparative anchor for the role of CbiA (F008).
- **Orthologous host genes:** Not applicable (no host disease gene).

15. Model Organisms

- **Mouse models:** *B. miyamotoi* infects laboratory mice; **SCID (severe combined immunodeficient) mice** were pivotal in demonstrating that Vmp segment conversion is

antibody-independent while clearance is antibody-dependent (F003). A dedicated laboratory mouse model to study BMD has been reported (PMID: 42367759).

- **Model utility:** immunocompetent vs immunodeficient mice dissect innate (complement/CbiA) vs adaptive (antibody/Vmp) contributions — recapitulating the human immunocompetent-vs-immunocompromised severity dichotomy.
- **In vitro:** *B. miyamotoi* is cultivable in modified Kelly-Pettenkofer medium, enabling complement-resistance and CbiA functional assays (PMID: 25189195, PMID: 28331202).
- **Heterologous expression:** ectopic *cbiA* in serum-sensitive *B. garinii* confers serum resistance — a gain-of-function validation (F008).
- **Limitations:** murine models may not fully capture human CNS disease; tick-transmission dynamics require the *Ixodes* vector.

Mechanistic Model / Interpretation

The unifying model of BMD is a **two-tier immune-evasion cascade** in which the outcome is set by the balance between bacterial evasion and host humoral immunity:

Tier	Effector	Immune arm evaded	Consequence	Evidence
Tier 1 (innate)	CbiA binds Factor H; blocks C3-convertase, reduces MAC	Complement (innate)	Survives serum killing → high spirochetemia	F008 (P 25104575 P 28331202 P 25189195)
Tier 2 (adaptive)	Vmp long-segment plasmid conversion	Antibody (adaptive)	Antigenic escape → relapse	F003 (P 41026790)
Resolution / severity switch	Specific antibodies clear all variants	—	Cure (immunocompetent) or CNS disease (immunocompromised)	F003, F005, F006

This model explains the full clinical spectrum from a single axis — **humoral competence**. Immunocompetent hosts eventually generate antibody waves that outpace Vmp switching and clear infection (self-limited, curable, no sequelae). Immunocompromised/B-cell-depleted hosts cannot clear the antigenically shifting population, permitting persistence and CNS invasion. CbiA-mediated complement resistance is the permissive upstream event that allows spirochetemia to reach the levels needed for both symptomatic disease and antigenic-variation-driven relapse.

Evidence Base

PMID	Contribution	Finding
34412488	Relapsing-fever phylogeny, <i>I. ricinus</i> complex transmission, rodent reservoirs	F001
33582142	1994 discovery, 2011 first human disease	F001
35858517	Transovarial + horizontal + transtadial transmission	F001
26053877	US case series: symptoms, 24% hospitalized, lab triad, doxycycline cure, serology kinetics	F002, F005, F006
26821411	Russian cohort: ~10% relapse, pre-treatment relapses	F002, F006
41026790	Vmp long-segment conversion; antibody-independent switching, antibody-dependent clearance (SCID)	F003
36113496	Meta-analysis: tick prevalence by species, 4.4% human seroprevalence	F004
31906865	Reference genome; 3 geographic populations	F004
38916722	CNS infection in immunocompromised	F005
36314925	Protein array (GlpQ, Vmps, flagellin)	F005
27626914	Pediatric case (age 5)	F007
41637958	Bbss-Bmiya coinfection at chance frequency only	F007
25104575	Reduced C3/C5/MAC deposition (complement resistance)	F008
28331202	CbiA identification and function	F008
25189195	Human complement resistance; cultivability	F008
41086691	Denmark seroprevalence gradient	F009
25356364	Netherlands occupational seroprevalence	F009
33370341	Low western-US seroprevalence	F009
24432595	Differential-diagnosis decision tree (~95% accuracy)	Diagnostics

Supporting surveillance literature: Pennsylvania statewide *I. scapularis* survey (PMID: 38686844), ArboNET DIN trends (PMID: 40907973), German tick-removal study (7.4% of *Borrelia*-positive ticks were *B. miyamotoi*; PMID: 31987819), Slovakia (PMID: 35094490), Kazakhstan (PMID: 39332111), Belgium (PMID: 39238018), and NY/Long Island clinical series (PMID: 32473652).

Limitations and Knowledge Gaps

1. **Non-genetic disease:** Sections 4 (Genetic/Molecular), 9 (Inheritance), and the genetic-testing portions of Section 10 are **not applicable** — BMD has no host genetic architecture. This is a definitive negative finding, not a data gap.
 2. **True incidence unknown:** Seroprevalence quantifies exposure, not clinical incidence. Underdiagnosis is likely because BMD is non-specific and overlaps clinically/geographically with Lyme disease.
 3. **Sex ratio and age distribution** of *clinical* cases are not well quantified across populations.
 4. **Diagnostic sensitivity limits:** Acute serology is insensitive (16%); PCR requires active spirochetemia; standardized commercial assays remain limited.
 5. **CNS disease is under-characterized** — reported almost exclusively in small case reports of immunocompromised patients; natural history and optimal CNS treatment duration are not established from trials.
 6. **No randomized treatment trials:** Doxycycline efficacy rests on case series and relapsing-fever precedent, not RCTs; optimal regimen/duration is empirically derived.
 7. **QoL and long-term outcomes** are inferred from "no chronic sequelae" observations rather than prospective cohorts.
 8. **Vaccine and prophylaxis** research is essentially absent.
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Proposed Follow-up Experiments / Actions

1. **Prospective incidence study** in high-endemicity regions using paired acute/convalescent PCR + GIpQ serology to convert seroprevalence into true clinical incidence and define age/sex distributions.

2. **Structural and functional dissection of CbiA** (AlphaFold model + Factor H co-crystal) to map the FH-binding interface and evaluate CbiA as a vaccine/therapeutic target; test *cbiA* knockout attenuation in the SCID vs immunocompetent mouse model.
3. **Longitudinal Vmp repertoire sequencing** during human/murine infection to quantify switching rate, cassette usage hierarchy, and correlation with relapse timing.
4. **Randomized/pragmatic treatment comparison** (doxycycline duration; ceftriaxone for CNS disease) to establish evidence-based regimens, including Jarisch–Herxheimer incidence.
5. **Improved point-of-care diagnostics:** multiplex PCR panels and next-generation serologic arrays (GlpQ + Vmp + flagellin) validated against paired sera; evaluate metagenomic sequencing for CNS disease.
6. **Immunocompromised-host registry** (especially rituximab-treated patients) to characterize CNS disease natural history, treatment response, and outcomes.
7. **Vector/reservoir surveillance integration** (ArboNET Tick Module expansion) to map acarological risk and guide clinician awareness where *B. miyamotoi* and *B. burgdorferi* co-occur.
8. **Vaccine feasibility assessment** targeting conserved surface antigens (CbiA, GlpQ), leveraging the observation that antibodies mediate clearance.

Report compiled from 9 confirmed findings and 35 reviewed papers across 5 investigative iterations. Evidence types: human clinical case series/cohorts, serosurveys, meta-analysis, in vitro microbiology, and mouse (including SCID) model studies.
